

Attention and the Economic Response to Immigration Enforcement

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Abstract

ICE enforcement actions reduce foot traffic in some immigrant neighborhoods and not in others. We study 51 single-worksite ICE enforcement events across 26 U.S. states from 2018 to 2026, estimating for each event a difference-in-differences model that compares foot traffic in heavily Latino-immigrant neighborhoods with otherwise similar neighborhoods before and after the event. The average response is close to zero, but we reject the hypothesis that the effect is common across events and estimate that about 70 percent of the cross-event variation in the estimates reflects true differences in the response rather than estimation noise. We then ask what explains this variation. A post-event spike in US Spanish-language news coverage of ICE predicts larger declines and accounts for roughly 30 percent of the true cross-event variation, while the number of arrests has no predictive power. English-language news and search-volume measures show the same pattern, and neither the heterogeneity nor its attention gradient is explained by differences in the validity of the design across events. An instrumental-variables design using news crowd-out from major foreign disasters isolates the effect of coverage on foot traffic for mid-visibility events. Once national attention is accounted for, local attention adds little. Commercial activity contracts modestly, while civic and institutional visits to schools, parks, museums, and related venues fall more.

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1 Introduction

Immigration enforcement can affect the lives of individuals beyond those who are detained. In part, this is because the community response may depend as much on the information environment surrounding an event as on the event itself. A workplace raid that goes largely unnoticed may have little effect beyond the worksite, whereas a smaller raid that becomes highly visible may lead residents to avoid public places. A long literature in economics and sociology documents these “chilling effects,” which can lead to reduced use of healthcare, schooling, and public benefits due to fear of detention [Watson, 2014, Asad, 2023]. The intensification of U.S. Immigration and Customs Enforcement (ICE) activity under the second Trump administration provides a real-time test. For example, Lester et al. [2026] document an 8–10 percent foot-traffic decline and a 20–25 percent spending decline in the most heavily Latin American-born neighborhoods of Los Angeles–Orange County in the weeks following ICE’s May 14, 2025 announcement of a major regional enforcement operation.

This raises a question that analysis of a single event cannot answer. Is the chilling effect due to the operation itself, or its visibility to the surrounding community? To the extent that observable operational scale is central, arrest counts should help explain which events produce larger community responses. If it is the broader *perception* of risk that is important, then the impact will depend on whether news of the event reaches people who were not directly targeted. An ICE raid clearly affects both margins at once. It detains a limited number of individuals at a specific site while also generating news coverage, social-media activity, rumor, and protest.

This paper uses cross-event variation to separate these channels. We assemble 51 ICE enforcement events from April 2018 through January 2026 spanning 26 states. The sample includes worksite raids from both the first and second Trump administrations including a small cluster of high-profile 2018–2019 actions and a much larger cluster of 2025–2026 worksite operations. For each event we estimate a distributed-lag difference-in-differences specification using Advan cellphone foot-traffic data, comparing Census Block Groups (CBGs) in the top decile of Latin-American foreign-born (FB) share to those in the bottom decile, and we extract a summary estimate of the average foot-traffic response.¹ Because an average across events can hide very different event-level responses, we first measure how much the true response actually varies. A random-effects decomposition rejects a common effect and attributes roughly 70 percent of the cross-event variation in the estimates to genuine differences rather than estimation noise, with the response to a new event ranging from a roughly 5 percent decline in some events to little or no response in others. We then ask which features of the event predict where in that range a neighborhood falls.

The strongest predictor of foot traffic is the public attention garnered by the event, not its operational scale. We exploit post-event spikes in Spanish-language news-volume queries for “ICE”, the agency name most salient to the population at risk, constructed from a collection of US Spanish-language outlets with daily coverage across the sample period. Events that receive more

¹This is the specification used by Lester et al. [2026] for their single LA event, which we implement across each of our events for comparison.

Spanish-language attention generate larger declines in foot traffic, and the same pattern appears in English-language news and Google search data. In a precision-weighted meta-regression, the Spanish-language attention spike accounts for roughly 30 percent of the true cross-event variation in responses, while arrest counts account for essentially none.

The pattern is robust to several checks. It holds when attention and arrests are included together, when noisier event estimates are down-weighted, when we account for pre-event media attention, and when we focus on within-event changes rather than levels. We also exploit news of major foreign disasters, which crowd out public attention on enforcement actions, as an instrument for the attention spike, with out-of-state U.S. disasters providing a complementary version. These sources of news pressure strongly predict the amount of attention given to enforcement events and are predetermined with respect to the operation itself. The one detectable imbalance, a mechanical clustering of disaster timing, is controlled directly. Because competing news shifts attention only for events whose coverage is sensitive to the news cycle, the instrument identifies an effect for mid-visibility events rather than for the largest, most visible operations. Balance checks show no systematic relationship with pre-event foot-traffic trends or placebo responses, and the estimates are robust to a variety of controls. Neither the heterogeneity nor its attention gradient is driven by differences in the validity of the event-study design across events. Measures of design quality are unrelated to attention, and both the dispersion and the gradient persist on subsamples restricted to events with no overlapping nearby actions and statistically flat pre-trends. The gradient is, if anything, stronger within the 2025–2026 enforcement wave considered on its own, so it does not reflect differences between the two enforcement eras.

Both national Google Trends and national Spanish-language news predict the local decline in foot traffic, while metro-level search and local Spanish-language news add little once national attention is accounted for. We also show that the community response is broad, extending beyond the immediate worksite and beyond the industry of the raided site. It is also present across the broad Latino community rather than concentrated among those at the highest personal deportation risk, since exposure measures restricted to non-citizens or to Mexican and Central American origin yield similar but weaker gradients.

We then decompose the response by venue type and show that both commercial activity and participation in public and institutional life are affected. We find that foot traffic around schools, parks, museums, and historical sites falls more than foot traffic at commercial sites (e.g., retail and food services), a pattern more consistent with a broad withdrawal from public life than with a localized demand shock.² The welfare cost of visible enforcement is therefore not limited to lost commercial activity but also includes reduced participation in civic life.

An important policy implication is that ambiguity itself can be costly. If residents respond to the attention garnered by enforcement as a signal of broader risk, then enforcement agencies and local officials can reduce harmful spillovers by communicating the scope of an operation, what it does and does not imply, and whether routine public and institutional settings remain outside the

²Religious organizations and hospitals are not present in the data, see Section 4.6.

action.

A broader economics literature uses changes in interior-enforcement intensity and federal enforcement programs, especially Secure Communities, to estimate the effects of immigration enforcement on safety-net participation, family decisions, childcare markets, healthcare use, and labor-market outcomes [Watson, 2014, Amuedo-Dorantes et al., 2018, 2020, Alsan and Yang, 2024, Ali et al., 2024, Herring and Barnow, 2025, East et al., 2023]. This work shows that enforcement can alter behavior and economic outcomes among people who are not themselves detained. A related sociology literature describes the everyday mobility and institutional-avoidance strategies of mixed-status Latino families [Asad, 2023]. Our paper differs by holding the enforcement event fixed and studying the public signal it generates. The object is not whether a larger enforcement regime changes behavior, but why two raids with similar operational severity can produce very different neighborhood responses.

Several recent papers are related. Amuedo-Dorantes and Antman [2022] show that ICE removals and raid awareness reduce immigrant labor-market engagement. Ciancio and García-Jimeno [2026] study consumption responses to learned enforcement risk, while De Balanzó et al. [2026], Hernandez [2026], and Cox and East [2026] estimate economic effects of the 2025 enforcement wave using spending, mobility, and labor-market data. These papers show that enforcement can move economic activity when risk becomes visible or learned. Our paper asks a different but complementary question. Conditional on a single-worksite enforcement event occurring, does the public signal around that event explain why some neighborhoods withdraw much more than others, holding the enforcement event fixed and varying its public visibility?

Our broader contribution is to the evidence on how public attention, more than an event’s operational scale, shapes the economic response to a salient risk. The COVID-mobility literature [Goolsbee and Syverson, 2021, Baker et al., 2020, Chetty et al., 2024] shows that perceived risk moved economic activity independently of formal mandates, but in that setting the risk and its visibility rise together. Enforcement events separate the two. The same operation produces a worksite disruption and a public signal, and the size of the community response tracks the signal rather than the operation. The relevant proxy for perceived risk is what the community is paying attention to, not what the operation looks like to the agency carrying it out. The same logic may extend to other salient but partly hidden risks, such as natural disasters, product recalls, or policy changes, whose economic footprint depends on how widely they are noticed, although we do not test that broader claim directly. Prior work instruments for or exploits quasi-experimental variation in enforcement itself. We instead hold the event fixed and instrument its visibility, using competing news pressure to move coverage of an enforcement action without changing the action. This adapts the news-pressure logic of Eisensee and Strömberg [2007] to isolate the attention component of a specific event, a strategy that applies wherever a discrete event competes for public attention and, to our knowledge, is new to the immigration-enforcement literature.

2 Data

The analysis combines the event inventory with outcome, exposure, and attention data. The outcome and exposure sources are cellphone foot traffic, SafeGraph consumer spending, and American Community Survey (ACS) demographics. The attention and news-pressure sources are Google Trends, Mediablast US Spanish-language news volume, GDELT 2.0 English-language news volume, Global Disaster Alert and Coordination System (GDACS) disaster alerts, and Nielsen Designated Market Area (DMA) definitions.

2.1 Event inventory

We compile a list of single-worksite ICE enforcement events from news sources, agency press releases, and litigation records. The inventory covers two distinct enforcement waves from April 2018 through August 2019 and January 2025 through January 2026 (the second term’s worksite operations).³ The inventory restricts attention to events that satisfy four criteria. First, the event date must fall within these two windows. Second, the event must correspond to a specific worksite or commercial address. Third, the worksite must be in one of 26 states for which we have CBG-level Latin American foreign-born data from the ACS. Finally, the worksite lat/lon must be geocoded either to the named business address or, when that is not feasible, to the city or county centroid.⁴ These screens identify 55 single-worksite ICE-led events. Four have insufficient catchment POI coverage to estimate the event-study specification, leaving a main analysis sample of 51 events spanning 26 states. Table 1 summarizes the event-level variables by enforcement wave. The 2018–2019 events report more arrests on average but generate much smaller news spikes than the 2025–2026 events, which motivates the era-robustness check in Section 4.1.

³The first Trump term’s worksite operations included Bean Station TN, Salem OH Fresh Mark, Mt. Pleasant IA Precast, Allen TX CVE Technology, Sumner TX Load Trail, and the August 2019 Mississippi poultry operation, which hit seven plants across five companies on a single day and which we include as a single Morton-MS-anchored compound event.

⁴Four events are explicitly coded as city-centroid locations, one additional event uses the earlier city-query fallback, and one event uses a county-level fallback.

Table 1: Event-level summary statistics, by enforcement wave

	Full	2025–2026	2018–2019
Arrests reported at worksite	70 (128)	46 (89)	215 (218)
Spanish-ICE news spike	0.0298 (0.0242)	0.0337 (0.0235)	0.0054 (0.0100)
English-GDELT news spike	0.0146 (0.0145)	0.0163 (0.0148)	0.0036 (0.0027)
Late-window estimate (log pts)	-0.008 (0.047)	-0.009 (0.047)	-0.003 (0.046)
High-FB CBGs in catchment	118 (167)	130 (175)	43 (69)
Low-FB CBGs in catchment	134 (165)	143 (172)	84 (100)
Events	51	44	7

Notes: Cells report the cross-event mean with the standard deviation in parentheses, for the full 51-event sample and separately by enforcement wave. News spikes are the post-event peak above the pre-event baseline in the share of stories mentioning the term (Section 2.6). The late-window estimate is the per-event weeks-4–8 differential in log foot traffic. High-FB and low-FB CBGs are the top- and bottom-decile Latin-American-foreign-born block groups in each event’s catchment. Arrests are missing for two events.

A public event inventory could be selected toward visible raids. No complete administrative list of single-worksite ICE enforcement actions is publicly available for these two windows, so we built the inventory from several discovery channels rather than national news alone, including agency releases, litigation records, immigration-policy trackers, state and local reporting, and prior event compilations. The resulting inventory includes a low-attention tail. Of the 55 single-worksite ICE-led events, 32 are coded medium or low publicity and 29 have an English-GDELT news spike at or below 0.01 (Section 2.6). The four POI-coverage exclusions include three medium-publicity events and one high-publicity event. These facts do not establish a complete universe of obscure raids, but they make it unlikely that the attention gradient is mechanically generated by retaining only highly visible events. Appendix Tables 8, 9, and 10 report the sample flow, event enumeration, and source-type audit.

2.2 Foot traffic

We obtain weekly visitor counts at point-of-interest (POI) level from Advan Research’s *Foot Traffic/Weekly Patterns Plus* dataset [Advan Research, 2025], derived from anonymized opt-in mobile devices. Each POI is geocoded to a latitude and longitude, assigned a NAICS code, and assigned to its CBG. For each event, we collect all Advan POIs within 50 km of the worksite anchor and assemble a $\text{POI} \times \text{week}$ panel spanning ± 12 weeks of the event date.

The resulting multi-event panel contains approximately 69 million POI-week observations and 3.3 million distinct POI–event pairs across 1.5 million unique POIs. Within each event’s catchment, we aggregate POI visits to CBG-week totals and restrict to CBGs containing at least 10 POIs. This restriction concentrates the analysis on retail-active CBGs.

2.3 Consumer spending

To measure spending, we use SafeGraph’s *SpendPatterns* dataset [SafeGraph, 2022], which reports monthly POI-level aggregates of consumer spending from a large opt-in financial-institution panel. Each row contains an array of daily spending values that we aggregate to weeks, producing a $\text{POI} \times \text{week}$ spending panel comparable to the foot-traffic data. We then aggregate POI-week to CBG-week using the same catchment definitions as the foot-traffic build.

2.4 ACS exposure measure

We define treatment as CBGs in the top decile of share of residents born in Latin America, which we designate as “high exposure”, while CBGs in the bottom decile serve as the comparison group. We use the 2018–2022 5-year ACS and calculate the foreign-born share from Latin America at the tract level and allocate it to all CBGs within each tract.

The deciles are computed within each event’s catchment, so the threshold for “top decile” varies across events. In heavily-Latino metros, the top-decile cutoff is roughly 32 percent foreign-born from Latin America, while in less-Latino metros the threshold is much lower in absolute terms. This within-event normalization means our cross-event comparison is between “most-Latino” and “least-Latino” CBGs in each respective metro.

2.5 Google Trends salience

For each event, we measure public salience using weekly Google Trends search interest for three keywords (“ICE raid”, “redada”, “deportación”) at three geographies: the U.S. national level, the event’s state, and its Nielsen Designated Market Area (DMA). Trends rescales each query within a geography and timeframe, so we use within-geography pre-vs-post changes rather than cross-geography levels. The pre-event baseline window is the four-week average over $\tau \in [-8, -5]$ weeks relative to the event Monday. We restrict Trends measures to the 2024–2026 enforcement wave because cross-era rescaling is noisy for these low-baseline search terms. DMA-level Trends are available for a 30-DMA subset.

2.6 News-volume series

We use two daily news-volume series spanning both enforcement waves on a consistent scale. The primary series measures attention in the Spanish-language news environment that the affected immigrant population consumes. A complementary series measures attention in the English-language US news supply.

Mediacloud US Spanish-language news. We query the Mediacloud Online News Archive [Media Cloud, 2025] for daily story counts in its curated US Spanish-language collection of 407 sources, including national, state, and local outlets. The headline query is the noun “ICE” restricted to this collection. We normalize by the daily total of all stories in the collection to produce a

share-of-coverage time series that is comparable across years. We also pull complementary Spanish keywords and construct a state-local Spanish-news series by restricting the same query to sources tagged as published in the event’s home state.

GDELT English-language US news. As a corroborating measure constructed on a different platform, language, and methodology, we use the GDELT 2.0 Global Knowledge Graph project’s Document API [Leetaru and Schrod, 2013]. We query the API for daily article counts matching the phrase “ICE raid” restricted to US English-language news sources, in normalized “Volume Intensity” units. The result is a single national daily time series spanning January 2018 through March 2026. GDELT closely tracks the Spanish-language Mediachannel series at the per-event spike level.

2.7 Competing-news instruments

To construct news-pressure instruments, we use three predetermined event calendars. The preferred instrument uses the Global Disaster Alert and Coordination System (GDACS) event feed [Global Disaster Alert and Coordination System, 2026]. We obtain red-alert sudden disasters, including earthquakes, tropical cyclones, floods, volcanic eruptions, and wildfires, that occur outside the United States, Latin America, and the Caribbean. The Latin America and Caribbean exclusion is deliberate because homeland shocks could directly affect Latino immigrant communities through family ties, remittances, or Spanish-language news demand rather than only through national news-cycle competition. For each ICE event, the instrument counts the number of days in the post-event window that overlap with the first seven days of one of these foreign-disaster alerts.

We also construct a complementary news-pressure measure using a hand-coded calendar of major U.S. natural disasters in states other than the ICE event state. The foreign- and U.S.-disaster instruments identify different margins of news-cycle access, so we interpret the IV estimates as local effects for events whose attention is moved by the corresponding source of competing news pressure.

3 Empirical Strategy

3.1 Per-event distributed-lag DiD

For each event e in our 51-event sample (single-worksite ICE enforcement events from April 2018 through January 2026, see Section 2.1), we estimate the same distributed-lag difference-in-differences model:

$$\log(1 + y_{c,t}^e) = \alpha_c^e + \lambda_t^e + \sum_{\tau \neq -1} \gamma_\tau^e (\text{HighFB}_c^e \times \mathbf{1}[t - T_0^e = \tau]) + \varepsilon_{c,t}^e, \quad (1)$$

where $y_{c,t}^e$ is weekly visitor count (foot traffic) at CBG c in week t , restricted to the catchment of event e . T_0^e is the Monday of the week containing event e ’s date. The dummy HighFB_c^e equals

one for CBGs in the top decile of share Latin American foreign-born within event e ’s catchment and zero for CBGs in the bottom decile (middle deciles are dropped). α_c^e and λ_t^e are CBG and calendar-week fixed effects, both event-specific because Equation (1) is estimated separately for each event. Standard errors are clustered at the CBG level. The event-time window is $\tau \in [-8, +8]$ weeks.

3.2 Identifying variation

The design has two steps. First, Equation (1) uses within-event variation between high-FB and low-FB CBGs in the same catchment. CBG fixed effects absorb time-invariant differences in population, retail density, and baseline foot traffic. Calendar-week fixed effects absorb shocks common to all CBGs in the catchment in a given week. The event estimate is therefore the differential foot-traffic response of heavily Latin-American-foreign-born CBGs relative to low-FB CBGs in the same metro and calendar weeks. Second, we use the 51 event estimates and ask whether cross-event differences in the attention spike predict cross-event differences in the estimated response.

The design does not require raids to be unanticipated, and some enforcement initiatives are clearly partially anticipated. In the initial specifications, the question is whether the media coverage generated by each event explains why otherwise similar raids produce different neighborhood responses. In a later section we implement an instrumental-variables design to isolate plausibly exogenous variation in the media coverage that is generated by the surrounding news cycle. In addition, Section 4.2 reports specifications that add pre-event media coverage as a control.

3.3 Information-shock and event severity measures

To test whether the chilling response is driven by information or by operational scale, we construct two categories of cross-event measures. The headline information-shock measure is the post-event spike in the Mediacloud US Spanish-language news-volume series for the query “ICE” (Section 2.6), measured as the share of stories in the 407-source US Spanish-language collection that mention the term. We compute the spike as the peak over $\tau \in [0, 8]$ minus the baseline over $\tau \in [-8, -5]$.

We complement the headline measure with three corroborating measures: the analogous GDELT national US English-language news-volume spike for “ICE raid,” the national Google Trends spike for “ICE raid,” and the DMA-level Google Trends spike for the local media market in which the event occurred. The two Trends measures are available on the 2024–2026 subsample and are defined as within-geography pre-vs-post changes, since Google Trends rescales each query within a geography and timeframe.

With respect to the severity of the enforcement event, we use two measures of underlying operational scale. These are the count of arrests reported by ICE at the worksite, Arrests_e , and the same measure in log form, $\log(1 + \text{Arrests}_e)$, to accommodate the right-skewed distribution. The arrests measure has coverage of 49 of 51 fitted events.⁵

⁵The two events with missing arrest counts in our inventory are Newark NJ (an unnamed business) and Baldwin County AL (an unnamed construction site).

3.4 Horse-race specification

To assess whether attention rather than severity predicts chilling, we run a cross-event OLS regression of the per-event late-window coefficient on a measure of attention and, in the severity specification, the log-arrests control:

$$\beta_{\text{late}}^e = \pi_0 + \pi_S \text{Spike}_e + \pi_A \log(1 + \text{Arrests}_e) + u_e, \quad (2)$$

where Spike_e is the post-event spike in one of the four information-shock measures defined above. Heteroskedasticity-robust (HC1) standard errors are reported. The columns of Table 3 report the headline Mediacloud Spanish-ICE specification alone and with the arrests control on the full 51-event panel, a standalone arrests specification, and single-regressor specifications using English-GDELT and national Google Trends as corroborating measures on the same windows. The Trends specification uses the 2024–2026 subsample. Comparing the single-regressor estimates across sources shows whether the same attention gradient appears outside the headline Spanish-language measure.

3.5 News-pressure instruments

The OLS horse race treats realized post-event attention as the explanatory variable. To isolate variation in attention generated by the surrounding news cycle, we use predetermined competing-news shocks as instruments. The preferred instrument is major foreign disaster news pressure. It counts days in the post-event window that overlap with the first seven days of a GDACS red-alert sudden disaster outside the United States, Latin America, and the Caribbean. The exclusion of Latin America and the Caribbean is intended to remove shocks that could directly affect Latino immigrant communities through homeland ties rather than through news-cycle competition only. The identifying assumption is that these shocks move enforcement action media coverage without directly changing the relative foot-traffic response in high-FB versus low-FB neighborhoods. Section 4.4 and Appendix C report the corresponding balance and robustness checks.

We also use out-of-state U.S. disasters as a complementary instrument. This measure captures domestic disaster-news pressure in states other than the ICE event state. The foreign- and U.S.-disaster instruments identify different local margins of news-cycle access. The IV estimates should therefore be read as local effects for events whose public attention is shifted by the corresponding source of competing news pressure.

4 Results

We begin with the event-level estimates from the distributed-lag design and use OLS to ask which event features predict larger neighborhood responses. We then implement the instrumental-variables strategy using variation in news volume from major foreign disasters, with out-of-state U.S. disasters as a complementary margin. These news shocks crowd out the volume of news

associated with enforcement actions.

4.1 Heterogeneity in the chilling response

The average neighborhood response to an enforcement event is close to zero, but the response varies far more across events than sampling error alone can explain. Across the 51 events the late-window (weeks 4 through 8) estimates average -0.008 log points and are not distinguishable from zero on average, yet a test of effect homogeneity is decisively rejected (Cochran’s $Q = 165$ on 50 degrees of freedom, $p < 10^{-13}$). A random-effects decomposition attributes about 70 percent of the cross-event variance in the estimates to true differences in the underlying response rather than to estimation noise ($I^2 = 69.7$ percent), with an estimated standard deviation of true effects of $\tau = 0.029$ log points. The implied 95 percent prediction interval for the response to a new event spans $[-0.057, +0.047]$ log points, about a 5.5 percent decline to a 4.8 percent increase. Table 2 reports the decomposition and Figure 1 plots the event estimates against the pooled mean and prediction interval. The positive and negative estimates split roughly as the near-zero mean and the estimated dispersion imply. The random-effects model puts about 43 percent of true effects above zero, in line with the 22 of 51 positive point estimates in the sample. The sign is ordered by attention. Negative estimates concentrate among high-attention events and positive estimates among low-attention events (Appendix Figure 2). The positive cases are therefore events where little public attention produced little chilling, rather than evidence that enforcement raises local activity.

This dispersion is structured by public attention. A random-effects meta-regression of the per-event estimate on the standardized post-event Spanish-language ICE news spike explains roughly 30 percent of the true between-event variance, and a one standard deviation larger spike predicts a 1.8 percent larger foot-traffic decline (-0.018 log points, $p < 0.001$ with Knapp-Hartung inference, Table 2, Panel B). Adding the log arrest count leaves the attention coefficient essentially unchanged, while the arrest coefficient is small and statistically insignificant.⁶ The English-language GDELT news spike has the same sign and significance and explains about 19 percent of the variance. The gradient survives controlling for operation type and sector and is robust to plausible corrections for measurement error in arrests, making it unlikely that the attention gradient simply proxies for operational scale (Appendix Table 16).

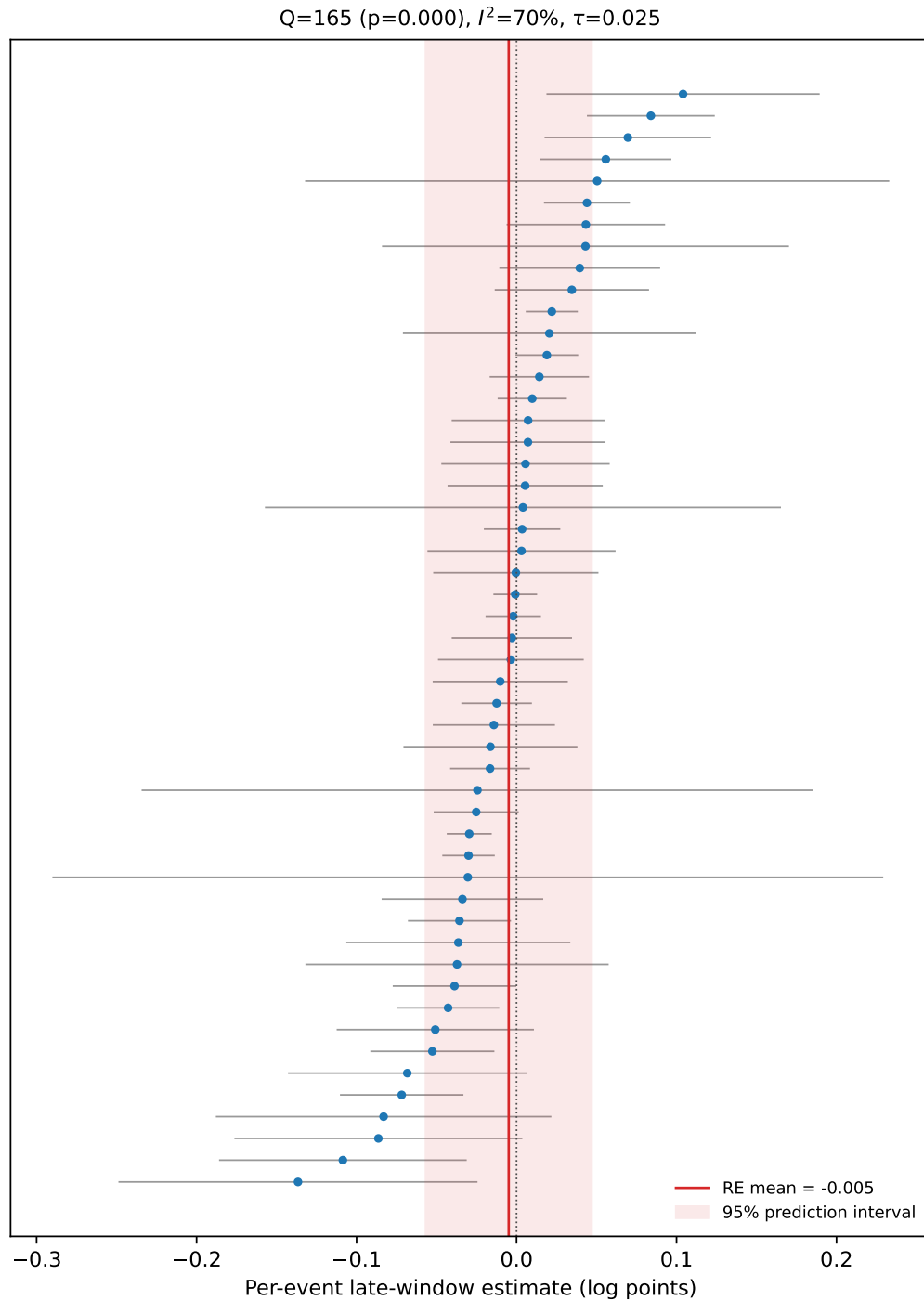
⁶Arrests are a coarse proxy for operational scale. A small arrest coefficient need not imply that no dimension of operational scale matters, only that the dimension captured by arrest counts does not predict the response.

Table 2: Direct estimation of cross-event heterogeneity

<i>Panel A: Heterogeneity of the late-window response</i>	
Events	51
Cochran’s Q	164.9
Q p -value	0.0000
I^2	69.7%
τ (REML, log points)	0.0287
τ (DL, log points)	0.0254
RE pooled mean	-0.0049
95% CI	[-0.0145, 0.0047]
95% prediction interval	[-0.0568, 0.0470]
<i>Panel B: Meta-regression on attention (per SD)</i>	
Spanish-ICE spike	-0.0184
(KH SE)	(0.0051)
KH p	0.0006
R^2_{meta}	30.1%
Spanish-ICE arrests	-0.0192
(KH SE)	(0.0053)
R^2_{meta}	25.1%
English-GDELT spike	-0.0154
(KH SE)	(0.0050)
R^2_{meta}	18.5%

Notes: Random-effects meta-analysis of the 51 per-event late-window (weeks 4–8) estimates, using the correct standard error of the late-window average (SE of the linear combination $c'\beta$ from the CBG-clustered covariance matrix, not the mean of the weekly SEs). τ is the estimated standard deviation of true effects across events. The 95% prediction interval is the Higgins–Thompson–Spiegelhalter interval for the true effect of a new event. Panel B reports random-effects meta-regressions of the per-event estimate on standardized attention spikes with Knapp–Hartung adjusted inference; R^2_{meta} is the share of between-event variance (τ^2) explained, with τ^2 estimated by DerSimonian–Laird for both the intercept-only and meta-regression models on the same sample. Spikes are standardized to the 51-event SD; the arrests specification has $N = 49$.

Figure 1: Per-event late-window estimates with the random-effects pooled mean and prediction interval



Notes. Each marker is one of the 51 fitted events, ordered by point estimate, with whiskers at ± 1.96 times the standard error of the late-window average (the standard error of the linear combination of weekly coefficients, clustered at the CBG level). The vertical line is the random-effects pooled mean and the shaded band is the 95 percent prediction interval for the true effect of a new event.

The heterogeneity is not an artifact of the event-study design varying across contexts. Per-event design quality, measured by the flatness of pre-event trends, the number of block groups in the catchment, the precision of the estimate, and exposure to overlapping nearby events, is uncorrelated with the attention spike, and the attention gradient is unchanged when these design characteristics are added as controls (Appendix Table 14). Restricting the sample to the 34 events with no overlapping-catchment contamination, and further to the 23 events with both no contamination and statistically flat pre-trends, leaves the measured heterogeneity intact, with I^2 of 61 and 63 percent, and preserves the attention gradient, although precision falls in the smallest subsample. The gradient is also a within-wave phenomenon rather than an artifact of the difference between the 2018–2019 and 2025–2026 enforcement eras. Estimated on the 2025–2026 wave alone, which contains 44 of the 51 events, it is if anything stronger (Appendix Table 15). The pooled event-time representation of the gradient (Appendix Figure 3) is descriptive rather than identifying. Its late-window estimate is sensitive to modest departures from parallel pre-trends, with a relative-magnitudes breakdown value near 0.1 (Appendix Table 13), so the cross-event design and the instrumental-variables estimates below, not the pooled time path, carry the identification.

4.2 The information-shock horse race

Table 3 reports the main horse-race result.⁷ Columns (1)–(2) anchor on the Mediacloud US Spanish-language news-volume spike for the query “ICE” (407-source Spanish-language collection, Section 2.6). Column (3) reports the standalone arrests specification. Columns (4)–(5) replace the headline measure with two corroborating attention proxies built on different platforms, the English-language GDELT national news-volume spike and the US national Google Trends spike for “ICE raid.”

⁷The horse-race and instrumental-variables coefficients are scaled per unit of the unstandardized attention spike, so they are larger in magnitude than the per-standard-deviation meta-regression coefficient in Section 4.1.

Table 3: Information shock horse race. Late-window estimate on attention and severity

	(1) Spa. ICE	(2) + arr.	(3) Arr. only	(4) GDELT only	(5) Trends only
MC Spa. ICE spike	−0.9193*** (0.2559)	−0.9987*** (0.2649)			
English GDELT spike				−1.3382*** (0.4382)	
National Trends spike $\times 10^{-3}$					−0.5254*** (0.1576)
$\log(1 + \text{Arrests}_e)$		−0.0053 (0.0042)	−0.0028 (0.0049)		
Constant	+0.0193** (0.0096)	+0.0386** (0.0184)	+0.0015 (0.0172)	+0.0114 (0.0088)	+0.0110 (0.0086)
N	51	49	49	51	44
R^2	0.227	0.260	0.006	0.173	0.215

Notes. OLS regressions of the per-event late-window estimate on the post-event attention spike (peak $[\tau \in [0, 8]]$ minus baseline $[\tau \in [-8, -5]]$, event-week aggregation) and $\log(1 + \text{Arrests}_e)$. The headline measure in columns (1)–(2) is the Mediacloud US Spanish-language news ICE-spike (407-source collection, share-of-coverage). Column (3) reports the standalone severity specification. Columns (4)–(5) report the same single-regressor attention specification with English-GDELT (full 51-event panel) and US national Google Trends ‘ICE raid’ (44-event 2024–2026 subsample) as corroborating attention measures. The headline measure closely comoves with the corroborating measures, so the single-regressor attention columns should be read as cross-source robustness rather than independent channels. Heteroskedasticity-robust (HC1) standard errors in parentheses. Stars denote * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In short, a larger Spanish-ICE spike predicts a larger decline in foot traffic. Adding $\log(1 + \text{Arrests}_e)$ leaves the Spanish-ICE coefficient essentially unchanged, while arrests alone explain very little of the cross-event variation. English-GDELT and national Google Trends show the same negative association with foot traffic.

Several robustness checks support this interpretation. Spanish-language news, English-language news, and national search measures move together across events. The post-event attention result is not explained by public news attention in the four weeks before the raid. Precision-weighted estimates, event-level bootstrap intervals, pre-event coefficients, and placebo estimates give the same conclusion. Appendix B reports these checks, along with an early-window table showing that the attention gradient builds over the first eight post-event weeks.

4.3 National vs. local attention

We next ask whether the primary effect of media coverage on foot traffic is national or local. Table 4 reports the levels version of the same-source national-vs.-DMA Google Trends horse race for the 2024–2026 subsample. This comparison holds the search platform and query fixed and varies only the geography of measured attention.

Table 4: Local vs. National Coverage of Enforcement Events

	(1) Nat.	(2) + arr.	(3) + DMA	(4) DMA	(5) DMA + arr.
National Trends spike $\text{Spike}_e^{\text{Nat}}$	-0.000525*** (0.000158)	-0.000564*** (0.000167)	-0.000621*** (0.000228)		
DMA Trends spike $\text{Spike}_e^{\text{DMA}}$			+0.000074 (0.000215)	-0.000326* (0.000174)	-0.000330* (0.000184)
$\log(1 + \text{Arrests}_e)$		-0.001275 (0.005263)			+0.000589 (0.006522)
Constant	+0.010989 (0.008648)	+0.015012 (0.017686)	+0.012604 (0.008953)	+0.006794 (0.008799)	+0.005279 (0.021078)
N	44	42	41	41	39
R^2	0.215	0.232	0.239	0.085	0.079

Notes: OLS regressions of the per-event late-window estimate on national and DMA-level Google Trends spikes for “ICE raid” and, in columns (2) and (5), $\log(1 + \text{Arrests}_e)$. Both Trends measures are post-event spikes, defined as peak search interest over $\tau \in [0, 8]$ minus the baseline average over $\tau \in [-8, -5]$. The Google Trends sample is restricted to the 2024–2026 events for which weekly search data are internally consistent. The DMA measure is missing for a small set of events whose media markets have insufficient search volume. Heteroskedasticity-robust (HC1) standard errors in parentheses. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

National Trends predicts larger declines in foot traffic and remains negative when DMA Trends is included. DMA Trends has the expected negative sign when entered alone, but it adds little once national attention is accounted for. Reported arrests remain null in the same exercise.

4.4 Instrumenting attention with news pressure

The relationship between media coverage and foot traffic raises the concern that because both are measured after the same enforcement action, they may be confounded. In particular, a larger neighborhood withdrawal, or the protest activity it draws, could itself attract more coverage, so the cross-event association may overstate the causal effect for the most visible events. The instrument addresses this simultaneity for events whose coverage is sensitive to the news cycle. We address this with an instrumental-variables strategy following the news-pressure logic of [Eisensee and Strömberg \[2007\]](#). In short, the idea is that predetermined national news-cycle conditions can shift attention to an enforcement event without altering the operation itself.

The preferred instrument is major foreign disaster news pressure. It counts the number of days in the post-event window $\tau \in [0, 8]$ that fall within the first seven days of a GDACS red-alert sudden disaster outside the United States, Latin America, and the Caribbean. These disasters are predetermined with respect to the enforcement event and to varying degrees they occupy national news attention. Because the outcome is a within-catchment differential between heavily Latino-immigrant neighborhoods and otherwise similar neighborhoods, a national news shock that moves attention equally in both groups nets out of this differential, so the residual threat is a home-region disaster that differentially affects the treated population. We therefore exclude Latin America and the Caribbean, where shocks could directly affect Latino immigrant communities through family

ties, remittances, or Spanish-language news demand. We also report out-of-state U.S. disasters as a complementary margin of domestic disaster-news pressure. These instruments need not identify the same local average treatment effect. Each isolates events whose enforcement attention is moved by a particular source of competing news pressure.

Table 5: OLS and IV estimates of the effect of ICE media coverage on foot traffic

	(1) OLS	(2) IV: Foreign disasters	(3) IV: US disasters (cross-state)	(4) Both IVs (over-id)
<i>Panel A: Mediacloud Spanish-ICE spike (endogenous)</i>				
Coverage coefficient	−0.919*** (0.256)	−1.059*** (0.338)	−0.911 (0.796)	−1.052*** (0.340)
<i>t</i> -statistic	−3.59	−3.13	−1.14	−3.09
First-stage <i>F</i> on instruments	—	52.66	28.44	25.78
Sargan over-id (<i>p</i>)	—	—	—	0.830
<i>Panel B: English-GDELT spike (endogenous, corroborating)</i>				
Coverage coefficient	−1.338*** (0.438)	−1.699*** (0.565)	−2.272 (1.779)	−1.682*** (0.563)
<i>t</i> -statistic	−3.05	−3.01	−1.28	−2.99
First-stage <i>F</i> on instruments	—	53.12	12.85	26.07
Sargan over-id (<i>p</i>)	—	—	—	0.748
<i>N</i>	51	51	51	51

Notes: Each panel reports the coefficient on the post-event news-attention spike in a regression of per-event β_{late}^e on attention, with no other covariates. Column (1) is OLS. Columns (2)–(4) are 2SLS specifications using predetermined news-pressure instruments. The foreign-disaster instrument is the number of days in the post-event window $\tau \in [0, 8]$ that fall within the first seven days of a GDACS red-alert sudden disaster outside the United States, Latin America, and the Caribbean. The US-disaster instrument counts major natural disasters in a different US state from the ICE event. Column (4) uses both instruments jointly. First-stage statistics are robust Wald *F* tests for the excluded instruments. Heteroskedasticity-robust standard errors in parentheses. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 5 reports the IV estimates against the OLS reference. Foreign-disaster pressure has the expected crowd-out sign in the first stage, and Table 5 reports first-stage *F* statistics for each specification. The second-stage estimates are similar to, and if anything slightly larger in magnitude than, the OLS estimates, consistent with classical measurement error in the attention spike attenuating the OLS association toward zero. The foreign-disaster IV is strong on its own, and the overidentified specification using both disaster instruments gives a similar result, although the U.S.-disaster instrument is uninformative in isolation. Reported arrests do not predict the foot-traffic response in any specification, so the instrument cannot transmit to the outcome through an arrest channel. The estimate is also unchanged when arrests are included as a control. The IV is consistent with a causal interpretation of the effect of media coverage on foot traffic for mid-visibility events at the margin of the news cycle. These are not the largest, most visible operations. Appendix C reports the first-stage coefficients, robustness checks, and balance checks.

4.5 Mechanisms

The evidence so far shows that media attention predicts the size of the neighborhood response to enforcement actions. We next ask what kind of response that attention produces. If direct disruption is the main channel, the attention effect should be strongest close to the worksite and in the same broad sector as the raided employer. If perceived risk is the main channel, the response should extend across a wider geography and across sectors.

Table 6: Mechanisms

Sample	Events	Mean estimate	Coef. on Spanish-ICE spike
<i>Panel A. Geographic reach, mutually exclusive distance rings</i>			
0–5 km from worksite	12	+0.031	–0.022 (0.017)
5–15 km from worksite	29	–0.013	–0.026*** (0.007)
15–30 km from worksite	36	+0.002	–0.023* (0.012)
30–50 km from worksite	40	–0.009	–0.009 (0.006)
<i>Panel B. Same-sector and broad-sector activity</i>			
Same broad sector as raided worksite	20	+0.003	+0.012 (0.017)
Other sectors	49	–0.011	–0.022*** (0.007)
Consumer-facing POIs	51	–0.007	–0.018*** (0.006)
Nonconsumer POIs	48	–0.003	–0.029*** (0.008)

Notes. Each row reports a cross-event OLS regression of a subgroup-specific late-window estimate on the standardized Mediacloud Spanish-ICE attention spike. Standard errors are heteroskedasticity-robust and shown in parentheses. Panel A defines high-FB and low-FB CBGs once within the full 50 km event catchment and then estimates the event-study separately by distance ring. Panel B estimates the event-study separately by POI bucket. Same-sector POIs have the same broad NAICS-2 code as the raided worksite. Consumer-facing POIs are NAICS-2 codes 44, 45, 72, and 81. Stars denote * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A of Table 6 estimates the event-study separately by distance from the worksite. The inner-ring estimates are noisy because only 12 events have enough high- and low-FB CBGs within 5 km, and the innermost band is slightly positive but imprecise. Even so, the negative response is not confined to the immediate worksite area. It appears in the 5–15 km and 15–30 km bands and remains negative, although smaller, in the 30–50 km band.

Panel B asks whether the response is concentrated in the raided sector. Same-sector estimates are available for only 20 events, largely because agricultural, industrial, and construction worksites often have few visitor-facing POIs in the same broad sector. In this sample, the same-sector

estimate is not negative. In contrast, other-sector POIs show a negative gradient close to the headline estimate. Consumer-facing POIs show the expected negative gradient, but nonconsumer POIs do as well. The response therefore does not look like a same-sector demand shock or a purely retail and restaurant channel.

Tests of the timing of news coverage point in the same direction. Weekly news peaks, national search, and an eight-week English-news area-under-the-curve measure all predict the response, while one-day peaks and simple days-above-threshold measures are weaker. Taken together, the mechanism evidence is most consistent with a broad perceived-risk response among heavily Latino-immigrant neighborhoods.

4.6 Civic vs. Commercial Response

We can also identify the broad activity types that are most affected by the enforcement actions. We re-estimate Equation (1) separately for two POI bins. The “civic/community” bin includes NAICS-3 codes 712 and 611, which cover parks, museums, and educational services. The “commercial” bin includes NAICS-2 codes 44, 45, and 72, which cover retail and food services. We omit NAICS-3 = 813 (religious organizations and civic associations) from the civic bin because the category is heavily contaminated by community-bank branches rather than civic organizations.

Table 7: Civic vs. commercial response, with inference

	N	Mean	(SE)	t
Civic estimate	12	−0.120**	(0.045)	−2.67
Commercial estimate	12	−0.010	(0.009)	−1.03
Civic − commercial	12	−0.111**	(0.040)	−2.79
High-attention events	6	−0.060	(0.044)	
Low-attention events	6	−0.162	(0.063)	

Notes: Sample is the 12 events with both a civic (NAICS-3 712 and 611: parks, museums, educational services) and a commercial (NAICS-2 44, 45, 72: retail and food services) per-event estimate. Each row reports a one-sample t -test across events of the indicated per-event estimate; the differential is computed within event before averaging. Inference is at the cross-event level and does not down-weight noisier event estimates. The attention split uses the Mediacloud Spanish-ICE spike. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Twelve events have both a civic and a commercial estimate. Across these events, civic visits in heavily-Latino CBGs fall by an average of 0.120 log points, statistically distinguishable from zero ($p = 0.02$), while the commercial decline is smaller and not distinguishable from zero. The within-event civic-minus-commercial differential averages −0.111 log points and is also statistically significant ($p = 0.02$; Table 7). The activity that falls most sharply is therefore institutional contact at schools, parks, and museums, rather than purely commercial activity. The civic sample is small,

and we cannot determine whether the civic-commercial gap is itself moderated by attention. The point estimates run opposite to that hypothesis, with a wider gap in low-attention events, but the two attention subgroups contain only six events each and are not statistically distinguishable. We therefore report the civic result as an average contrast and not as an attention-moderated effect. Appendix E reports the venue-coverage limits.

4.7 Spending

We re-estimate Equation (1) using $\log(1 + \text{weekly spending})$ from the rebuilt daily-unpacked Safe-Graph SpendPatterns panel. Forty-three events have enough high- and low-FB CBG coverage to estimate the event-level spending response. Individual-event spending estimates are much noisier than foot-traffic estimates, and the aggregate cross-event spending estimate is statistically indistinguishable from zero. Despite the noise, spending responses are more negative in events with larger Spanish-ICE and English-GDELT news spikes, and the decline is concentrated in high-attention events. Above-median-attention events show a differential spending decline of about 3 percent in heavily Latino-immigrant neighborhoods over the eight following weeks, while below-median-attention events show no decline. Because the panel records only card spending at covered merchants and the estimate is the excess decline relative to comparison neighborhoods in the same catchment, it understates the total economic cost of the response.

5 Conclusion

This paper asks why ICE enforcement events generate large neighborhood responses in some cases and little response in others. Across 51 single-worksite events in 26 states, the average response is near zero, but the effect varies far more across events than chance allows, and about 70 percent of the cross-event variation reflects true differences in the response rather than estimation noise. That variation is structured by public attention. Events that become visible in national Spanish-language news, English-language news, and web search generate larger declines in foot traffic in heavily Latino-immigrant neighborhoods, and attention explains roughly 30 percent of the cross-event variation while arrest counts explain essentially none.

The interpretation is that enforcement operates through an information environment as well as through direct detention. A raid disrupts a particular worksite, but it also sends a public signal about enforcement risk. Whether that signal reaches people outside the worksite depends on news coverage, search activity, protest, rumor, and the broader attention cycle around the event. The instrumental-variables design based on major foreign disasters supports this reading for events at the margin of news-cycle access, with out-of-state U.S. disasters providing a complementary margin. The design does not say that all enforcement news shocks are identical, or that the largest events in the sample are governed by the same margin, but it shows that predetermined variation in news pressure moves enforcement coverage and that this movement predicts the neighborhood response.

The venue decomposition also changes the welfare interpretation. Commercial activity contracts

modestly, and the commercial response is the part most naturally connected to retail-spending projections. The larger response appears in civic and institutional venues that include schools, parks, museums, and related public-facing institutions. Since the foot traffic panel does not cover churches or hospitals, this result is best read as evidence of institutional withdrawal in the venues we observe, with an important measurement gap in other civic settings. The cost of enforcement visibility is therefore not only lost transactions. It is also reduced participation in public and institutional life.

Importantly, the most visible events, including the Los Angeles cluster, are unlikely to be IV compliers because their attention spikes are large enough to persist through any reasonable amount of competing coverage. The spending leg of the analysis also has limited power. The venue decomposition is silent on foot traffic associated with religious congregations or healthcare venues, both of which are central to the January 2025 sensitive-locations rescission and to the broader immigration-fear literature. The civic estimate is consequently restricted to schools, parks, and museums. Additionally, the event inventory is built from public sources rather than a complete administrative list.

Taken together, the results show that the consequences of enforcement depend not only on the operation itself but also on the information environment it creates. Media coverage can turn a localized enforcement action into a broader signal of risk, and communities respond by reducing activity well beyond the worksite, the raided sector, and ordinary commercial activity. The same lesson applies wherever a salient event’s economic footprint runs through public attention rather than the event itself, and it points to a practical lever for enforcement. Agencies and local officials cannot control the news cycle, but they can limit avoidable spillovers by communicating clearly about an operation’s scope and limits and whether everyday civic and institutional settings fall outside it. This lever is most relevant for the mid-visibility events whose coverage is still sensitive to the news cycle, since for the most prominent operations attention is already saturated.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI’s ChatGPT and Codex to assist with coding, analysis checks, proofreading, and manuscript editing. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the article.

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A Single-worksite event inventory

The compiled single-worksite inventory contains 55 ICE-led events satisfying the event-date, geography, and geocoding screens described in Section 2.1. Table 9 lists the 51 events in the main analysis sample and the four single-worksite events that lack sufficient catchment POI coverage to fit the per-event distributed-lag DiD. Inclusion in the main sample requires at least 10 high-FB and 10 low-FB CBGs with sufficient POI coverage in the catchment. The four catchment-coverage exclusions are data-availability-driven rather than attention-correlated.

Table 8: Sample-flow from full event inventory to estimation subsamples

Definition	Events
Single-worksite ICE-led events satisfying inventory screens	55
– Single-worksite ICE-led but no estimable late-window estimate (insufficient catchment POI coverage) ^a	–4
Main analysis sample (single-worksite ICE-led with estimable late-window estimate)	51
<i>Estimation subsamples:</i>	
Events with reported arrest counts (severity test)	49
Events in 2024–2026 subsample (Google Trends national)	44
Events with DMA-level Google Trends coverage	41
Events with at least one in-state Mediacloud Spanish-language source	31
Events with estimable per-event civic estimate	11

Notes: “Events” is the unit. The 51-event panel is the main analysis sample for the per-event distributed-lag DiD and the headline cross-event horse race. Smaller numbers below reflect availability of specific second-stage covariates rather than re-estimation. The Mediacloud Spanish-language national series and the English-GDELT series are defined for all 51 events. Trends, DMA Trends, and in-state Spanish sources cover smaller subsamples. The civic estimate requires sufficient civic-NAICS POI density in the high-FB CBGs of an event’s catchment.

Table 9: Single-worksite ICE-led event inventory

Date	State	Event / worksite	Arrests
Main sample ($n = 51$)			
2018-04-05	TN	Southeastern Provision	97
2018-05-09	IA	MPC Enterprises (Midwest Precast Concrete)	32
2018-06-05	OH	Corso’s Flower & Garden Center (Sandusky + Castalia)	114
2018-06-19	OH	Fresh Mark	146
2018-08-28	TX	Load Trail LLC	159
2019-04-03	TX	CVE Technology Group	280
2019-08-07	MS	Mississippi poultry operation (7 plants, 5 companies)	680

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(Table 9 continued)

Date	State	Event / worksite	Arrests
2025-01-23	NJ	Newark	
2025-02-03	PA	Philadelphia	7
2025-02-12	TX	Los Fresnos	2
2025-02-18	NY	Tupper Lake Pine Mill	9
2025-02-26	MS	3 J Underground LLC	7
2025-02-26	MS	Gulf Coast Prestress Partners Ltd	18
2025-02-26	NJ	North Bergen	16
2025-02-27	PA	Jumbo Meat Market	4
2025-03-27	CA	Powder & Protective Coatings	5
2025-04-02	WA	Mt. Baker Roofing	37
2025-04-22	VT	Berkshire	8
2025-04-27	CO	Colorado Springs	100
2025-05-02	NY	Lynn-Ette & Sons Farms	14
2025-05-13	FL	Wildwood	33
2025-05-20	WA	Eagle Beverage	17
2025-05-21	AL		
2025-05-27	LA	Mirabeau Water Garden stormwater project	20
2025-05-29	FL	Perla	100
2025-05-30	CA	Los Angeles	36
2025-05-30	CA	Buono Forchetta	3
2025-06-04	TX	Brownsville	25
2025-06-05	PA	Wyoming Valley Pallets Inc.	4
2025-06-06	CA	Los Angeles	44
2025-06-10	CA	Oxnard	35
2025-06-10	NE	Glenn Valley Foods	76
2025-06-11	PA	Five10 Flats restoration jobsite	17
2025-06-17	LA	Delta Downs Racetrack	84
2025-07-02	TX	Groomer's Seafood	12
2025-07-08	NJ	Alba Wine and Spirits	20
2025-07-10	CA	Glass House Farms	361
2025-07-10	TX	Taco Ole restaurant	18
2025-07-16	PA	Super Gigante	14
2025-08-20	NJ	Edison	29
2025-08-23	CT	Optimo Car Wash	7
2025-09-02	MO	Golden Apple Buffet	12
2025-09-04	GA	HL-GA Battery Company	475

continued on next page

(Table 9 continued)

Date	State	Event / worksite	Arrests
2025-09-04	NY	Nutrition Bar Confectioners	57
2025-10-09	OH	Pancho’s Tacos	5
2025-10-15	CT	Hamden	7
2025-10-19	ID	La Catedral Arena	105
2025-10-30	OR	Woodburn	35
2025-10-30	LA	Barrois Welding Services shipyard	25
2025-11-04	MA	Allston Car Wash	9
2025-11-14	AL	Nori Thai and Sushi	18
<i>Excluded: insufficient catchment POI coverage ($n = 4$)</i>			
2018-08-08	NE	O’Neill NE agricultural operation (3 sites)	
2025-06-04	NM	Outlook Dairy Farms	11
2026-01-15	MN	D.R. Horton	
2026-01-15	TX	El Paso	38

Table 10: Source-type audit for the single-worksite inventory

Primary cited source type	Inventory	Main	Med./low	Low spike
Prior compilation or tracker	23	22	11	14
State or local news	15	15	9	7
Agency or public record	10	9	8	4
National news	3	3	0	1
Business or address verification	2	2	2	1
Trade/compliance tracker	2	0	2	2
Total	55	51	32	29

Notes: The table classifies the primary citation recorded in the event inventory. “Inventory events” are the 55 single-worksite ICE-led events satisfying the screens in Section 2.1. “Main sample” counts the 51 events with sufficient catchment POI coverage to estimate the event-level distributed-lag DiD. “Low news spike” denotes events with an English-GDELT post-event spike at or below 0.01. The table is an audit of the public-source inventory rather than a claim that the inventory is an administrative universe of all obscure raids.

B Additional result diagnostics

B.1 News measures, inference, and local attention

The cross-language and cross-source validity of the headline measure can be quantified directly. The Mediacloud Spanish-ICE event spike closely tracks the English-GDELT ICE-raid spike across the full event panel and closely tracks the US national Google Trends ICE-raid spike where Trends data are available. The English-GDELT spike also tracks Spanish-language search spikes for “redada” and “deportación” in the 2024–2026 subsample. Event-level ICE news cycles propagate consistently through English news supply, Spanish news supply, and English and Spanish search demand on essentially the same schedule.

The choice of search term within the Mediacloud Spanish corpus matters. The headline uses the noun “ICE,” which has by far the cleanest event-specific signal. Alternative phrasings such as “agentes de ICE” and “redada migratoria” give the same negative sign but weaker statistical power. “Deportación” is essentially null, consistent with the term tracking the Spanish-language political-policy news cycle rather than event-specific raid coverage.

Because the headline selects one measure from this family, we correct for multiple testing across the five news-volume measures. Table 11 reports Holm and Romano-Wolf family-wise adjusted p -values. The Spanish-ICE coefficient remains significant after both corrections (Holm $p = 0.003$, Romano-Wolf $p = 0.007$), as does the English-GDELT measure, while the weaker Spanish query variants do not. The Romano-Wolf adjustment, which accounts for the strong positive correlation across the measures, is smaller than the conservative Holm bound.

Table 11: Multiple-testing correction across attention measures

Attention measure	Coef	t	p	Holm p	RW p
Spanish “ICE” (headline)	-0.0184	-3.65	<0.001	0.003	0.007
English-GDELT national	-0.0154	-3.06	0.004	0.014	0.024
Spanish “agentes de ICE”	-0.0095	-1.74	0.088	0.177	0.212
Spanish “redada migratoria”	-0.0104	-1.97	0.054	0.162	0.177
Spanish “deportación”	+0.0019	+0.35	0.726	0.726	0.735

Notes: Each row is a separate random-effects meta-regression of the per-event late-window estimate on the standardized attention measure, with Knapp–Hartung inference and inverse-variance weights (same estimator as Table 2); $N = 51$. The family is the set of news-volume measures discussed in this appendix. “Holm p ” is the Holm step-down family-wise adjusted p -value, which controls the family-wise error rate under arbitrary dependence. “RW p ” is the Romano–Wolf step-down adjusted p -value from a recentered pairs bootstrap (2,000 resamples of events, fixed seed), which accounts for the strong positive correlation across measures and is therefore less conservative than Holm. The headline Spanish “ICE” coefficient remains significant after both corrections.

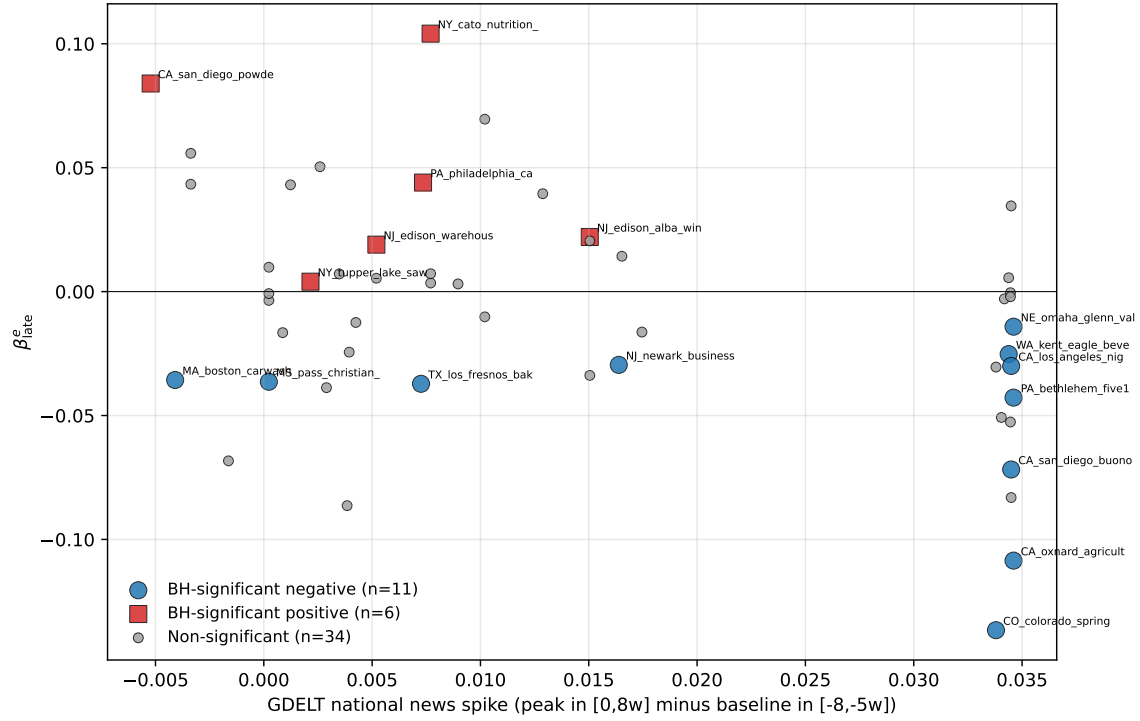
The generated-regressor concern does not drive the result. In inverse-variance-weighted WLS, where each event is weighted by the inverse of its squared standard error from the distributed-lag

DiD, the Spanish-ICE coefficient is smaller than the OLS estimate but remains statistically significant. Adding arrests leaves the result intact. An event-level bootstrap on the OLS specification also excludes zero.

The post-event attention measure also does not appear to be standing in for anticipatory media attention. For each news-volume series we construct an immediate pre-event spike, defined as the peak over $\tau \in [-4, -1]$ minus the same $\tau \in [-8, -5]$ baseline used in the headline post-event spike. The immediate pre-event Spanish-ICE spike is uncorrelated with both the late-window estimate and the average pre-event foot-traffic coefficient. Adding it to the headline horse race leaves the post-event Spanish-ICE coefficient unchanged, and the English-GDELT measure gives the same pattern.

We also construct an event-local Spanish-news series by restricting the Mediacloud query to outlets tagged as published in the event’s home state. The local-Spanish spike moves in the same direction as the national headline, but its predictive power is weaker and it adds little once national Spanish-language attention is included. This mirrors the same-platform national-vs.-DMA contrast within Google Trends. The predictive content of local-language local-news attention lies primarily in the part that tracks the national signal.

Figure 2: Per-event late-window estimate vs English-GDELT national news spike, by BH- $q < 0.05$ significance group



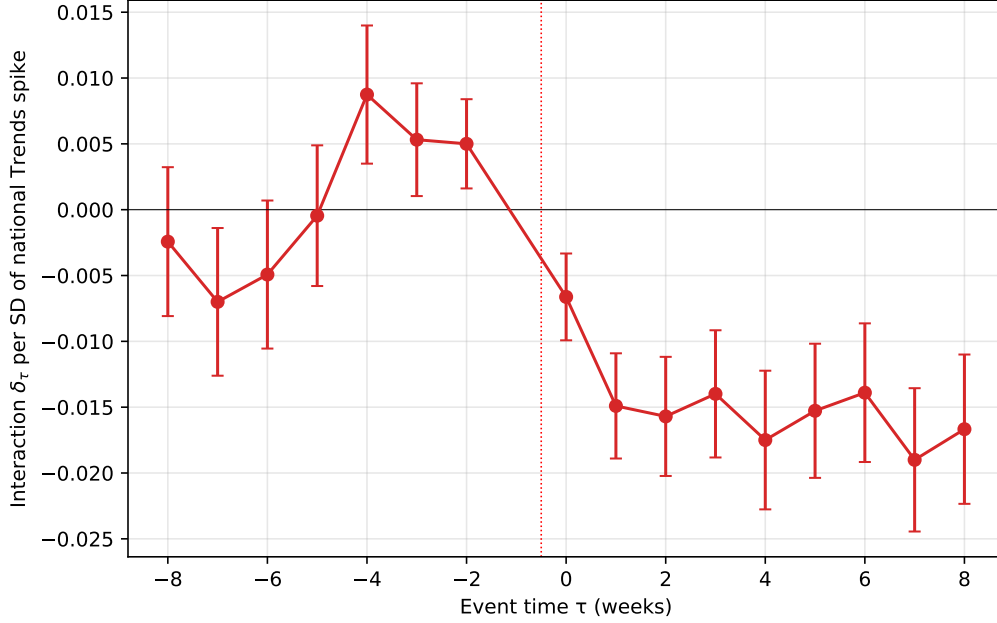
Notes. Each marker is one of the 51 fitted events. Blue circles denote events with a BH-corrected per-event significance of $q < 0.05$ and a negative late-window estimate (chilling). Red squares denote BH-significant events with a non-negative late-window estimate. Grey dots denote non-significant events. English-GDELT spikes are higher among the negative-significant group than the positive-significant group. The equivalent figure built on the headline Mediacloud Spanish-ICE measure gives the same qualitative pattern.

Table 12: Early-window robustness

	Weeks 0–2	Weeks 1–3	Weeks 4–8
<i>Panel A. Event-estimate distribution</i>			
Mean estimate	−0.001	−0.004	−0.008
Median estimate	0.000	−0.003	−0.003
Share negative	0.51	0.55	0.57
<i>Panel B. Cross-event attention gradient</i>			
Pearson r , Spanish-ICE spike	−0.263	−0.375	−0.477
OLS coef., Spanish-ICE spike	−0.323*	−0.524***	−0.919***
	(0.174)	(0.191)	(0.256)
OLS coef., Spanish-ICE spike, with arrests	−0.388**	−0.592***	−0.999***
	(0.181)	(0.202)	(0.265)
<i>Panel C. Corroborating English-news measure</i>			
Pearson r , English-GDELT spike	−0.227	−0.333	−0.416
OLS coef., English-GDELT spike	−0.466	−0.774**	−1.338***
	(0.302)	(0.333)	(0.438)
N	51	51	51

Notes. Each column summarizes the same per-event distributed-lag DiD path over a different post-event window. Week 0 is the calendar week containing the enforcement event and can include pre-event days. The weeks 1–3 window excludes the event week. The weeks 4–8 column is the headline persistent-response window used in Table 3. OLS rows report single-regressor HC1 standard errors in parentheses, except for the “with arrests” row, which also includes $\log(1 + \text{Arrests}_e)$ as a control. The arrests-control regressions use 49 events. Stars denote * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure 3: Pooled event-time attention interaction



Notes. Estimates from a single pooled regression on the stacked CBG-week panel.

$\log(1 + y) = \alpha_{c,e} + \lambda_{w,e} + \sum_{\tau \neq -1} [\gamma_\tau + \delta_\tau \text{Spike}_e^z] \cdot \text{HighFB}_c^e \cdot \mathbf{1}[t - T_0^e = \tau] + \varepsilon$, where Spike_e^z is the standardized event-level national Google Trends spike. The figure plots δ_τ . Standard errors are clustered at the CBG level.

Whiskers are $\pm 1.96 \times \text{SE}$. Reference week is $\tau = -1$. $N = 204,510$ CBG-week observations. The pre-event coefficients are not perfectly flat, so the figure is treated as descriptive rather than as an identifying test.

Because the pre-event coefficients in Figure 3 are not flat, we treat this pooled time path as descriptive and quantify how sensitive it is to violations of parallel trends. Table 13 applies the relative-magnitudes sensitivity analysis of Rambachan and Roth [2023] to the late-window average of the pooled event study. The robust confidence set for the interaction δ_τ already includes zero once the allowed post-event deviation from parallel trends reaches roughly one-tenth of the largest pre-event deviation (a breakdown $\bar{M} \approx 0.1$). The pooled time path therefore does not support identification on its own. Identification instead rests on the cross-event meta-regression of Section 4.1, which the design-validity checks below show is not contaminated by differential pre-trends, and on the instrumental-variables estimates of Section 4.4.

Table 13: HonestDiD pre-trend sensitivity of the pooled event study

Robust 95% CI (late-window avg)	
<i>Panel A: Interaction δ_τ (attention gradient, Figure 3), est. = -0.0165</i>	
Parallel trends ($\bar{M} = 0$)	[-0.0212, -0.0118]
$\bar{M} = 0.1$	[-0.0277, -0.0057]
$\bar{M} = 0.2$	[-0.0356, 0.0021]
$\bar{M} = 0.3$	[-0.0441, 0.0103]
$\bar{M} = 0.4$	[-0.0479, 0.0187]
$\bar{M} = 0.5$	[-0.0479, 0.0272]
$\bar{M} = 1.0$	[-0.0479, 0.0479]
$\bar{M} = 1.5$	[-0.0479, 0.0479]
$\bar{M} = 2.0$	[-0.0479, 0.0479]
Breakdown $\bar{M}$	0.10
<i>Panel B: Main γ_τ (average response), est. = -0.0023</i>	
Parallel trends ($\bar{M} = 0$)	[-0.0076, 0.0029]
$\bar{M} = 0.1$	[-0.0110, 0.0063]
$\bar{M} = 0.2$	[-0.0156, 0.0109]
$\bar{M} = 0.3$	[-0.0207, 0.0161]
$\bar{M} = 0.4$	[-0.0261, 0.0215]
$\bar{M} = 0.5$	[-0.0315, 0.0268]
$\bar{M} = 1.0$	[-0.0535, 0.0535]
$\bar{M} = 1.5$	[-0.0535, 0.0535]
$\bar{M} = 2.0$	[-0.0535, 0.0535]
Breakdown $\bar{M}$	0.00

Notes: Rambachan–Roth (2023) relative-magnitudes sensitivity for the pooled event study (national Google Trends measure, 7 pre-periods, 9 post-periods). The target is the late-window average ($\tau \in [4, 8]$). $\bar{M}$ bounds the largest post-treatment deviation from parallel trends by $\bar{M}$ times the largest observed pre-treatment deviation; $\bar{M} = 0$ is the standard CI under exact parallel trends. The breakdown $\bar{M}$ is the largest value at which the robust 95% confidence set still excludes zero. Confidence sets use the conditional least-favorable method.

B.2 Clean-event robustness

We compute a contamination metric for each event. The metric is the number of other in-sample events whose dates fall within ± 8 weeks of event e and whose CBG catchments overlap with e 's. Of the 51 main-sample events, 34 have zero contaminating neighbors. Of those 34, 24 fall in the 2024–2026 subsample for which the pre-event state-Trends baseline is internally consistent (Section 2.5). On this subsample the saturation-test correlation between the late-window estimate and the pre-event state-Trends baseline strengthens slightly relative to the full 2024–2026 Trends sample, in the same direction and with the same statistical strength. The leave-one-out diagnostic for the headline Spanish-ICE horse race keeps the coefficient close to the full-sample point estimate when any single event is dropped. The corresponding English-GDELT leave-one-out diagnostic leads to

the same conclusion.

Table 14 addresses the concern that the cross-event heterogeneity reflects differences in the validity of the event-study design rather than differences in the underlying response. Panel A regresses each per-event measure of design quality, namely the joint F statistic for pre-event trends, the magnitude of the pre-event and placebo coefficients, the number of block groups in the catchment, the standard error of the estimate, and the contamination metric, on the standardized attention spike. None is significantly related to attention, so high- and low-attention events do not differ systematically in how clean their comparisons are. Panel B shows that the attention coefficient in the meta-regression is unchanged when these design measures are added as controls. Panel C recomputes the heterogeneity statistics and the attention gradient on progressively cleaner subsamples. Restricting to the 34 uncontaminated events, and then to the 23 events that are also free of detectable pre-trends, leaves I^2 near 60 percent and preserves the attention gradient, though statistical power falls in the smallest subsample.

Table 14: Design validity: per-event design quality is orthogonal to attention

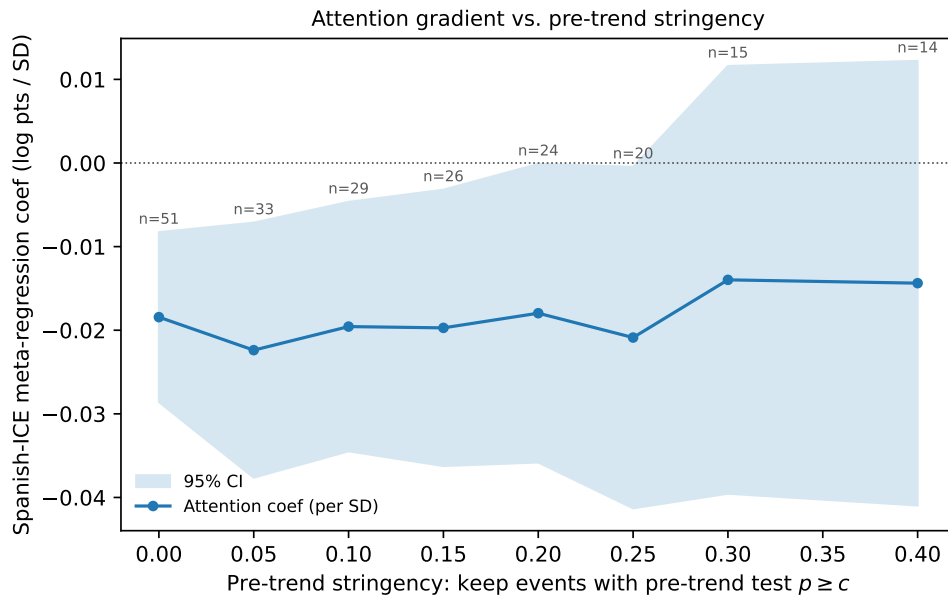
<i>Panel A: Design metric regressed on the standardized Spanish-ICE spike (HC1)</i>					
Design metric	N	slope/SD	t	p	R^2
Pre-trend joint F (non-flatness)	51	-217.6642	-1.04	0.301	0.047
pre-event coef	51	-0.0056	-1.31	0.191	0.045
placebo response	51	-0.0060	-1.25	0.210	0.035
# CBGs in catchment	51	-9.3960	-0.24	0.810	0.001
Per-event SE (imprecision)	51	-0.0005	-0.11	0.915	0.000
# contaminating events	51	0.1667	1.04	0.296	0.025
Frac. catchment shared	51	0.0256	0.51	0.611	0.004
<i>Panel B: Attention coefficient in the meta-regression, with vs. without design controls</i>					
	N	coef/SD	SE	t	p
Spanish-ICE spike (per SD), no controls	51	-0.0184	0.0051	-3.65	0.001
+ design controls	51	-0.0185	0.0054	-3.42	0.001
<i>Panel C: Heterogeneity on clean subsamples</i>					
Subsample	N	I^2	τ	spike p	R^2_{meta}
All events	51	69.7%	0.0287	0.001	30.1%
No contamination	34	61.1%	0.0309	0.027	14.8%
No contam. & flat pre-trends	23	62.8%	0.0301	0.051	16.6%

Notes: Panel A regresses each per-event design-quality metric on the standardized Spanish-ICE news spike with HC1 standard errors; a null slope means design quality does not vary systematically with attention. Panel B shows the random-effects meta-regression attention coefficient is unchanged when standardized design metrics (pre-trend F , # CBGs, # contaminating events) are added as controls. Panel C recomputes the heterogeneity statistics and the attention meta-regression on progressively cleaner subsamples (no overlapping-catchment contamination; additionally non-rejected flat pre-trends at $p > 0.10$). τ is in log points.

Figure 4 traces the attention gradient as the sample is restricted to events with progressively

flatter pre-trends. The meta-regression coefficient stays between -0.018 and -0.022 log points per standard deviation and remains negative and statistically significant as the pre-trend cutoff tightens from the full sample to the events that fail to reject flat trends at $p \geq 0.25$, a restriction that discards 60 percent of events. The confidence interval widens as the sample shrinks while the point estimate is stable. Only at the two strictest cutoffs, which retain about 14 events, does the coefficient attenuate to roughly -0.014 and lose significance, consistent with a loss of power rather than a change in the underlying relationship. The attention gradient is therefore not an artifact of the events whose pre-trends are least credible.

Figure 4: Attention gradient by pre-trend stringency



Notes. Each point is the random-effects meta-regression coefficient on the standardized Spanish-ICE attention spike, estimated on the subsample of events whose pre-event joint trend test fails to reject flat trends at $p \geq c$ (higher c requires flatter pre-trends). The spike is standardized once on the full 51-event sample, so the coefficient is in constant per-standard-deviation units. The shaded band is the 95 percent Knapp-Hartung confidence interval and labels give the number of events retained at each cutoff.

Table 15 reports the attention gradient estimated separately by enforcement wave. Within the 2025–2026 wave alone, which contains 44 of the 51 events, the gradient is at least as strong as in the full sample, so the cross-event relationship is not an artifact of the structural differences between the two enforcement eras. The 2018–2019 wave has only seven events, too few for inference, but its slope has the same sign.

Table 15: Attention gradient by enforcement wave

Sample	N	Spanish-ICE coef (per SD)	p	I^2
Full sample	51	−0.0184	0.001	69.7%
2025–2026 wave	44	−0.0199	0.001	73.1%
2018–2019 wave	7	−0.0149	n/a	n/a

Notes: Random-effects meta-regression of the per-event late-window estimate on the standardized Mediacloud Spanish-ICE spike, estimated separately by enforcement wave, with Knapp–Hartung inference. The spike is standardized within each sample. The 2018–2019 wave has only seven events, too few for inference, so its slope is reported descriptively and I^2 and p are omitted.

The gradient is also unlikely to be an artifact of mismeasured operational scale. Table 16 shows that the attention coefficient is essentially unchanged when arrests, operation type, and sector are added as controls, that attention dominates arrests in a per-standard-deviation horse race, and that the small arrest coefficient remains small relative to the attention coefficient even under generous corrections for measurement error in arrests. Arrests are the only continuous operational-scale measure the shock inventory records, so we control for it directly rather than building a scale index from a single variable, with operation type and sector capturing composition.

Table 16: Operational scale and the attention gradient

	Attn. coef	(KH SE)	p	R^2_{meta}	N
<i>Panel A: Attention coefficient as scale controls are added</i>					
Spanish-ICE spike	-0.0184	(0.0051)	0.0006	30.1%	51
+ log arrests	-0.0192	(0.0053)	0.0007	25.1%	49
+ arrests, type, sector	-0.0190	(0.0074)	0.0141	15.1%	49
English-GDELT spike	-0.0154	(0.0050)	0.0036	18.5%	51
+ log arrests	-0.0161	(0.0052)	0.0035	16.9%	49
<i>Panel B: Per-SD horse race, attention vs. log arrests</i>					
Attention spike (per SD)	-0.0192	(0.0053)	0.0007	25.1%	49
Log arrests (per SD)	-0.0040	(0.0055)	0.4675	—	—
<i>Panel C: Arrests de-attenuation (single-regressor errors-in-variables)</i>					
reliability $\rho = 1.0$ (no measurement error)	-0.0030	implied arrests coef / attention coef = 0.16			
reliability $\rho = 0.7$	-0.0042	implied arrests coef / attention coef = 0.23			
reliability $\rho = 0.5$	-0.0059	implied arrests coef / attention coef = 0.32			
reliability $\rho = 0.4$	-0.0074	implied arrests coef / attention coef = 0.40			

Notes: Random-effects meta-regressions of the per-event late-window estimate on standardized regressors, with Knapp–Hartung adjusted inference and inverse-variance weights from the clustered linear-combination standard errors (same estimator as Table 2). Arrests are the only continuous operational-scale measure recorded in the shock inventory: fines are unrecorded for every event, and operation end-dates are recorded for only 1 of 51 events (and reflect case resolution rather than operation length), so we control for arrests directly rather than constructing a scale index from one variable. Operation type/composition controls are worksite operation, sector agriculture, sector food, sector manufacturing, sector construction. Publicity is a visibility measure and is deliberately excluded from the scale block. Panel C reports the implied true single-regressor arrests coefficient b_{arr}/ρ under classical measurement error with reliability ρ ; even at $\rho = 0.5$ the implied coefficient is small relative to the attention coefficient. Attention coefficients are per standard deviation of the spike; the arrests specifications have $N = 49$.

The headline meta-regression is also robust to how its inference is conducted. Table 17 reports a randomization-inference p -value that permutes the attention spike across events ($p = 0.0006$), leave-one-out estimates that remain negative and significant in all 51 fits, and the heterogeneity standard deviation under three estimators (DerSimonian–Laird, REML, and Paule–Mandel), all between 0.025 and 0.031 log points. The gradient is not an artifact of the normal approximation underlying Knapp–Hartung inference, of a single influential event, or of the τ^2 estimator.

Table 17: Inference robustness for the attention meta-regression

<i>Panel A: Attention slope, alternative inference</i>	
Slope (per SD)	-0.0184
Knapp–Hartung p	0.0006
Randomization-inference p	0.0006
R^2_{meta}	30.1%
<i>Panel B: Leave-one-out and influence</i>	
LOO slope range	[-0.0202, -0.0161]
LOO fits significant at 5%	51 of 51
Most influential event (drop)	slope $\rightarrow$ -0.0161
<i>Panel C: Heterogeneity SD τ by estimator</i>	
τ (DerSimonian–Laird)	0.0254
τ (REML)	0.0287
τ (Paule–Mandel)	0.0314
I^2	69.7%

Notes: $N = 51$. Panel A compares the Knapp–Hartung p -value for the random-effects meta-regression slope (per-event late-window estimate on the standardized Spanish-ICE spike, inverse-variance weighted) with a randomization-inference p -value that permutes the spike across events 10,000 times and recomputes the slope; the permutation test makes no normality or large-sample assumption. Panel B drops each event in turn: the slope range, the number of leave-one-out fits that remain significant at 5%, and the slope when the single most influential event is dropped. Panel C reports the heterogeneity standard deviation τ under three estimators. The slope is negative and significant under permutation inference and in every leave-one-out fit, and τ is stable across estimators.

The gradient also does not reflect how events entered the inventory. Table 18 classifies each event by the type of its source citation. Events documented through news outlets carry higher attention spikes than those documented through official or other channels, but discovery source does not predict the per-event estimate, and the attention gradient is, if anything, stronger under discovery-source fixed effects.

Table 18: Discovery-source robustness

<i>Panel A: Attention spike and estimate by source type</i>			
Source type	Events	Mean spike	Mean β_{late}
news/web	32	0.0340	-0.0094
official	9	0.0349	+0.0050
other	10	0.0116	-0.0156
<i>Source $\rightarrow$ spike, joint F</i>		5.25	$p = 0.01$
<i>Source $\rightarrow \beta$, joint F</i>		1.63	$p = 0.21$
<i>Panel B: Attention gradient with source fixed effects</i>			
	Slope	(KH SE)	p
Baseline	-0.0184	(0.0051)	0.0006
+ discovery-source FE	-0.0228	(0.0051)	0.0001

Notes: $N = 51$. Each event is classified by the type of its inventory source citation (official government, news/web, advocacy/social, or other; categories smaller than 5 events are pooled into “other”). Panel A reports the mean attention spike and mean late-window estimate by source type and the joint F from regressing the standardized spike, and the estimate, on source dummies. Panel B is the random-effects attention meta-regression with and without discovery-source fixed effects. The source citation is a proxy for the discovery channel, not a measured discovery process; this is a selection probe, not an identifying control.

C Instrument robustness and balance

Table 19: First-stage relationship between news pressure and attention

	Mediacloud Spanish-ICE		English-GDELT	
	(1) Foreign	(2) US disaster	(3) Foreign	(4) US disaster
Foreign disaster days (std.)	-0.0168*** (0.0023)		-0.0105*** (0.0014)	
Cross-state disaster days (std.)		-0.0078*** (0.0015)		-0.0031*** (0.0009)
First-stage F	52.66	28.44	53.12	12.85
R^2	0.482	0.103	0.520	0.046
N	51	51	51	51

Notes: The dependent variable is the indicated post-event attention spike. Each column reports a single-instrument first stage with the instrument standardized. The foreign-disaster measure counts days in the ICE event’s post-event window that fall within the first seven days of a GDACS red-alert sudden disaster outside the United States, Latin America, and the Caribbean. The US-disaster measure counts major natural disasters in a different US state from the ICE event. Heteroskedasticity-robust standard errors in parentheses. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 20: Competing-news IV robustness: foreign disasters and controls

	(1) Foreign	(2) + sev./era	(3) + event ctrls	(4) Both IVs
Attention coef.	−1.059*** (0.338)	−1.123*** (0.341)	−0.993** (0.475)	−1.052*** (0.340)
<i>t</i> -statistic	−3.13	−3.29	−2.09	−3.09
First-stage <i>F</i>	52.66	77.83	24.80	25.78
Sargan over-id (<i>p</i>)	—	—	—	0.830
Foreign-disaster IV	Yes	Yes	Yes	Yes
US-disaster IV	No	No	No	Yes
log(1 + arrests) + 2025+	No	Yes	Yes	No
Publicity + sector ctrls	No	No	Yes	No
<i>N</i>	51	49	49	51

Notes: The dependent variable is per-event β_{late}^e and the endogenous regressor is the Mediacloud Spanish-ICE attention spike. Columns (1)–(3) use the foreign-disaster instrument alone. Column (4) uses the foreign-disaster and US-disaster instruments jointly. Severity/era controls are log(1 + arrests) and an indicator for events in 2025 or later. Event controls add the event-publicity flag and broad sector indicators for agriculture/dairy, restaurant/food, manufacturing/industrial, car wash, and construction. First-stage statistics are robust Wald *F* tests for the excluded instruments. Heteroskedasticity-robust standard errors in parentheses. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 20 shows that the foreign-disaster IV result is not an artifact of obvious event-level controls. Adding log(1 + arrests) and a second-Trump-era indicator leaves the estimate close to the headline foreign-disaster estimate. Adding event-publicity and broad-sector controls attenuates the estimate but preserves the sign and statistical strength. The overidentified specification with foreign disasters and out-of-state U.S. disasters also leads to the same conclusion.

Table 21: Balance check: foreign disaster news pressure and event characteristics

Covariate	<i>N</i>	Coef. on foreign disasters (cond.)	<i>p</i>
Log arrests	49	+0.296	0.034
Pre-event beta	51	−0.002	0.624
Backward placebo beta	51	+0.004	0.674
Cross-state disaster days (std.)	51	+0.486	0.003
Joint test of listed covariates	49	$F = 4.35$	0.006

Notes: Each row reports the coefficient on standardized foreign-disaster days in a regression of the listed predetermined event characteristic on foreign-disaster days, second-Trump-era, event-publicity, and broad-sector controls. The foreign-disaster measure counts days in the ICE event’s post-event window that fall within the first seven days of a GDACS red-alert sudden disaster outside the United States, Latin America, and the Caribbean. The joint test reverses the regression and asks whether the listed covariates jointly predict foreign-disaster days after the same era, publicity, and sector controls. Heteroskedasticity-robust inference is used. The table is a balance diagnostic, not a proof of exclusion.

Table 21 addresses the most direct timing concern. The question is whether foreign-disaster exposure is proxying for observable differences in which events occur when. The timing measure is related to the U.S.-disaster measure and to reported arrests, which is why we report the specification with both instruments and controls. It is not related to pre-event foot-traffic trends or the placebo response.

Table 22 pressure-tests the exclusion restriction directly. The restriction requires that foreign-disaster news pressure shift the neighborhood response only by crowding out coverage of the enforcement action. Because the outcome is a within-catchment differential, U.S.-wide disaster shocks difference out, and the main residual concern is a home-region channel in which a disaster in a country tied to the treated population affects those neighborhoods directly. The qualifying disasters are predominantly Asia-Pacific (73 percent), with the remainder elsewhere outside Latin America and the Caribbean, none with a material tie to the Mexican or Central-American-origin population. Restricting the instrument to Asia-Pacific disasters alone leaves the estimate close to the headline with a first-stage F of 47, and dropping the largest U.S. non-Latino diaspora origin countries preserves a significant estimate. The Africa-and-Europe-only slice has too few exposed events to support a first stage. The overidentified specification that adds U.S. cross-state disasters, which would violate the exclusion through a different channel, does not reject the Sargan test ($p = 0.83$). The instrument is unrelated to the pre-event coefficient and the placebo response, and it correlates with reported arrests and, mechanically, with the U.S.-disaster measure, which is why we control for arrests and report the two disaster instruments jointly.

Table 22: Exclusion-restriction pressure tests for the foreign-disaster IV

	Coef.	(SE)	First-stage F	Disasters
<i>Panel A: Geographic restriction of the instrument</i>				
Baseline (ex-LatAm/Caribbean)	−1.059***	(0.338)	52.7	113
Asia-Pacific only	−1.240***	(0.402)	46.6	83
Africa / Europe / MENA only	−0.520	(0.857)	5.6	27
Drop largest US (non-Latino) diaspora origins	−0.889**	(0.439)	25.8	69
<i>Panel B: Overidentification and balance</i>				
Foreign + US disasters (over-id)	−1.052***	(0.340)	25.8	
Sargan over-id p	0.830			
Balance joint F (covariates $\rightarrow$ instrument)	$F = 4.35, p = 0.006$			
OLS reference	−0.919***	(0.256)		

Notes: The endogenous regressor is the Mediacloud Spanish-ICE attention spike and the outcome is per-event β_{late}^e . Panel A re-estimates the foreign-disaster IV restricting the instrument to disasters by region of the affected country. Of the 113 qualifying red-alert non-Latin-American sudden disasters, 83 (73%) are in the Asia-Pacific and the remainder elsewhere outside Latin America and the Caribbean, none with a material tie to the Mexican or Central-American-origin population. The diaspora row drops the largest US non-Latino foreign-born origin countries (the Philippines, India, China, Vietnam, Korea). Panel B reports the overidentified specification using the foreign-disaster and US-disaster instruments jointly with its Sargan test, and the joint test that predetermined event characteristics predict the instrument. First-stage statistics are robust Wald F tests. Heteroskedasticity-robust standard errors. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 23 unpacks the joint balance test. The rejection is driven almost entirely by the mechanical correlation between foreign-disaster timing and the cross-state U.S.-disaster measure, which is the second instrument rather than a confounder of foot traffic; the predetermined design-quality covariates, the pre-event coefficient and the backward placebo, are each individually unrelated to the instrument. The foreign-disaster second-stage estimate is stable when these covariates are added as controls, moving from -1.06 to -1.04 , so the balance result reflects a controllable timing correlation rather than a channel that drives the estimate.

Table 23: IV balance: what drives the joint test, and conditional-IV survival

<i>Panel A: Loadings of predetermined covariates on the instrument (per SD)</i>				
	Coef	<i>t</i>	<i>p</i>	
Log arrests	+0.237	+1.26	0.208	
Pre-event β	-0.129	-1.14	0.255	
Backward placebo β	-0.081	-0.47	0.637	
Cross-state disaster days	+0.338	+3.05	0.002	
<i>Joint F (all four)</i>	4.35		0.006	
<i>Panel B: Foreign-disaster IV second stage, conditioning on the imbalance covariates</i>				
	Coef	(SE)	First-stage <i>F</i>	<i>N</i>
No controls (baseline)	-1.059	(0.338)	52.7	51
+ imbalance covariates	-1.056	(0.324)	62.0	49
+ imbalance, era, publicity	-1.040	(0.359)	92.4	49

Notes: Panel A regresses the standardized foreign-disaster instrument on the four predetermined covariates that enter the appendix joint balance test (Table 21), with the same era, publicity, and broad-sector controls; coefficients are per standard deviation of each covariate. The joint F reproduces the appendix value and identifies which covariate carries the rejection. Panel B re-fits the foreign-disaster 2SLS coverage-on-foot-traffic estimate, adding the predetermined covariates as controls (the cross-state disaster measure is the second instrument, not a control). The second-stage coefficient is stable across specifications, so the balance result reflects a controllable mechanical correlation rather than a violation that drives the estimate. The conditional specifications drop two events with missing arrest counts ($N = 49$). Heteroskedasticity-robust inference throughout.

D Exposure-measure robustness

The headline high-FB indicator uses share of Latin-American foreign-born population (ACS B05006) at the CBG level. This is a broad measure because it counts naturalized US citizens alongside non-citizens, all Latin-American national-origin groups alongside the highest-enforcement-risk Mexican and Central American populations, and all adults alongside the US-citizen children of foreign-born parents. To test whether the chilling response is concentrated in narrower subgroups most directly exposed to deportation risk, we refit the per-event distributed-lag DiD with three sharper exposure indicators. These are share Hispanic non-citizen adults, share foreign-born from Mexico, El Salvador, Guatemala, Honduras, or Nicaragua, and share US-citizen children under 18 with at least one foreign-born parent.

Table 24: Exposure-measure robustness, per-event estimate on Spanish-ICE spike by high-FB indicator

Exposure indicator	N	mean estimate	sd estimate	r vs base	coef	t	R^2
Latin-Amer. FB (baseline, B05006)	51	−0.011	0.044	+1.000	−0.92	−3.59	0.227
Hispanic non-citizen (B05003I)	51	−0.007	0.046	+0.440	−0.17	−0.41	0.006
Mex. + Cent. Amer. FB (B05006)	51	−0.006	0.043	+0.716	−0.58	−1.98	0.089
Mixed-status kids (B05009)	49	−0.001	0.040	+0.551	−0.29	−1.05	0.027

Notes: For each exposure indicator, we refit the per-event distributed-lag DiD (Equation 1) defining high-FB CBGs as the top-decile and low-FB CBGs as the bottom-decile within each event’s catchment by the indicated variable. “Mean estimate” and “sd estimate” are the cross-event distribution of the resulting late-window estimates. “ r vs base” is the cross-event correlation of the refined-indicator estimates with the baseline estimates. The last three columns report the single-regressor OLS of the late-window estimate on the Mediacloud Spanish-ICE spike with HC1 standard errors.

All three refinements produce the same negative sign as the baseline at smaller mean magnitudes and substantially weaker cross-event predictive power for the Spanish-ICE attention spike. The pattern suggests that the cross-event response operates at the broad community level rather than narrowly at the personal-deportation-risk level. This weakening is not mechanically driven by fewer POIs in the sharper top-decile cells. Top-decile POI density under the sharper indicators is within ± 6 percent of the baseline.

E Venue-coverage and civic-mechanism checks

Advan’s POI panel does not measurably capture two venue categories that are central to the sensitive-locations policy and to the broader immigration-fear literature. Religious organizations (NAICS-3 = 813) contain 4,025 POIs across the 26-state catchment universe, of which 3,817 (95 percent) are the Independent Community Bankers of America contamination documented in Appendix F. After stripping obvious mislabels, only a very small number of panel POIs correspond to real religious congregations. Healthcare venues are absent. NAICS-2 = 62 has zero POIs in the panel. The civic-versus-commercial decomposition therefore captures the schools and parks/museums response but is silent on churches and hospitals.

We also test whether the civic-vs.-commercial differential is itself moderated by the cross-event attention spike. With the English-GDELT spike, the within-event civic-minus-commercial differential is -0.153 in the low-spike group and -0.051 in the high-spike group. With the Mediacloud Spanish-ICE spike, the pattern is similar. The differential is -0.162 in the low-spike group and -0.060 in the high-spike group. The civic-vs.-commercial gap is wider in low-attention events than in high-attention events under both measures, the opposite of what an attention-moderated mechanism would predict.

F POI data-quality audits

F.1 Advan and SafeGraph POI classification

This section discusses non-trivial point-of-interest classification errors in the underlying SafeGraph POI universe. These include NAICS misclassification across sectors, POIs identified as active businesses that are in fact vacant or demolished structures, and visit-count series that are mathematically inconsistent with the POI’s physical footprint.

F.2 Geometric POI flags

We construct two independent quality flags for all Advan POIs in the catchments of every event in our sample, covering approximately 1.54 million distinct POIs across the event catchments used in the audit. Building-containment is tested against Microsoft Building Footprints, and freeway-proximity is tested against TIGER Primary and Secondary Roads. A POI receives the freeway-traffic flag if its lat/lon falls outside any building footprint and is within 50 meters of an interstate centerline. A POI receives the remote flag if it falls outside any building, more than 100 meters from the nearest building, and more than 200 meters from any S1100 or S1200 road.

Across the 26-state panel, 7,881 POIs receive the freeway-traffic flag (0.5 percent) and 20,324 receive the remote flag (1.3 percent), for a combined drop rate of about 1.8 percent. The flagged set catches the expected error mode. In California, the five most-visited flagged POIs are the Port of Los Angeles, the Port of Long Beach, an Irwindale aggregates quarry, the Long Beach port-adjacent industrial complex, and the Port of Los Angeles San Pedro terminal. The flagged POIs are disproportionately in low-Latino-foreign-born CBGs rather than high-Latino-foreign-born CBGs, so removing them shrinks the comparison group rather than the treatment group. Re-estimating the per-event late-window effect on the cleaned panel shifts the cross-event mean by 0.0004 log-points. Three of 51 events flip sign, all with absolute estimates below 0.01 in either direction. The mean absolute change per event is 0.0056 log-points. The information-shock finding does not depend on the small subset of POIs that fail the building-containment and freeway-proximity checks.

F.3 POI verification and sector recoding

We supplement the geometric check with hand-verification of 100 randomly sampled California POIs, stratified by Veridion firm-registry match quality. Of the 100, 96 correspond to a real business or commercial property at the listed address. The four exceptions are all residential single-family homes that Advan labels with a generic NAICS-category string, and all four fall in the no-Veridion-match tier where Advan errors should be concentrated. No POI in the sample was a demolished structure or a business that never existed.

A second verification of 500 California POIs stratified by NAICS-3 was designed to test whether classification noise could reshuffle the civic-versus-commercial buckets. The Advan-bucket $\times$ true-bucket cross-tabulation shows that the directional contamination that would erase the mechanism

finding is essentially zero. Only 1 of 200 civic-labeled POIs is actually commercial. The audit did identify a consequential bug in NAICS-3 = 813. In California, 110 of the 118 Advan POIs in NAICS-3 = 813 are labeled “Independent Community Bankers of America” but correspond to individual community-bank branch addresses. A further three NAICS-813 records are sports stadiums. We therefore drop NAICS-3 = 813 from the civic bin in Table 7. Including 813 attenuates the civic estimate toward zero by mixing in community-bank visitor activity whose own per-event response is closer to the commercial-bucket value.

F.4 Structural name-level checks

To move from spot-checks to panel-wide quantification, we conduct a verification exercise at the name-recurrence level. The 1.15M-POI 21-state catchment universe contains only 129,773 unique `location_name` values. The top 1,500 names cover 978,660 POIs. We classify each of the top names as a national chain with consistent NAICS, a generic NAICS-category-as-name label, a public-sector entity, a brand-at-host pattern in which a service brand sits inside a host retailer, or a trade-association-at-members pattern in which an association name is spread across member-business addresses.

The verification identifies 62 names with bulk-recodable structural miscoding, of which 44 require an actual NAICS-3 change. The 44 effective recoding rules cover 63,924 POIs (5.5 percent of the 21-state universe by POI count and 5.3 percent of the panel by visitor-week count). The largest pattern is money-transfer-agent POIs that live inside grocery and convenience-store hosts. For the civic-versus-commercial decomposition, however, the implication is limited. Only 148 POIs would leave the civic bucket under the recoding rules, and the audit identifies no name that would move POIs from civic to commercial or commercial to civic. Re-estimating the category decomposition after applying the full set of structural recoding rules moves the high-baseline civic estimate from -0.135 to -0.120 , while the commercial estimate moves from -0.001 to $+0.001$. The order-of-magnitude civic-versus-commercial gap is preserved.

F.5 Panel coverage

A separate concern is that businesses may be real but missing from the Advan POI universe altogether. If Advan systematically under-covers immigrant-serving businesses, the chilling regression would be attenuated downward in heavily Latino CBGs because the relevant POIs are not in the panel. We test this using Veridion as an external firm registry. For each 5-digit ZIP in the panel, we count Advan POIs and Veridion firms and compute the coverage ratio. Restricting to dense urban ZIPs and stratifying by foreign-born share quartile, the firm-weighted Advan/Veridion ratio is 0.28 in the lowest-FB quartile and 0.33 in the highest-FB quartile. The mild positive gradient does not support systematic undercoverage of immigrant neighborhoods.

Two Census establishment references corroborate the Veridion-based finding. At the county $\times$ NAICS-3 level, comparing Advan POIs to Census County Business Patterns, the establishment-weighted Advan/CBP coverage ratio rises monotonically from the lowest-FB quartile to the highest-

FB quartile. At the ZIP level, comparing against Census ZIP Business Patterns, the Advan/ZBP coverage ratio is essentially flat across foreign-born-share quartiles. Three independent external references therefore point the same direction. The panel-composition channel is a real concern in principle, but it is not the empirical source of the result.

F.6 Residential labels and visit-pattern mismatches

A further exercise targets residential single-family homes that Advan labels with a generic NAICS-category string. We construct a residence-mislabel flag combining a generic NAICS-category-as-name label with a strict-residential street suffix. The flag identifies 28,204 POIs (2.4 percent of the 21-state panel). A stronger version that also requires no Veridion firm within 100 meters identifies 17,456 POIs (1.5 percent). Spot-checks confirm high precision. Only 157 of the flagged residences fall in the civic bucket, and dropping them would shrink the civic bucket by less than 1 percent. The remainder of the flag concentrates in the commercial and other buckets and does not shift the commercial estimate off its near-zero baseline.

A final check addresses visit-count series that are inconsistent with what the POI claims to be, using only the visit data itself. For each POI we aggregate the Advan day-of-week and hour-of-day visit arrays over all of 2025, normalize each to a shape profile, and build a reference signature for every NAICS-3 with at least 30 well-measured POIs. A POI is flagged `flag_visit_pattern_mismatch` if both its day-of-week and hour-of-day profile correlate below 0.30 with its own sector’s reference signature. The flag identifies 75,963 POIs (6.6 percent of the panel). It is the noisiest quality check. The flagged set mixes genuine anomalies with legitimate businesses that keep atypical hours. We report it as a diagnostic rather than fold it into the headline cleaning, because at this rate and with non-trivial false positives it is too blunt to use as a panel filter. Its value is that it catches a class of error that no external firm registry can detect, a real address with a real NAICS label but an implausible visit series.